Executable Positive Geometry: An End-to-End Verification Tour from Bell Tests to the Two-Loop Amplituhedron, with Holographic Wedge Localization Measured on Superconducting Hardware

Jon Torrez*

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Abstract

We report a systematic, fully executable verification tour of the modern "physics from geometry" program, spanning three of its pillars: (i) quantum nonlocality and error correction, (ii) emergent geometry from entanglement, and (iii) positive geometries underlying scattering amplitudes. Every result is either derived symbolically in exact arithmetic, computed numerically without approximation, or measured on IBM superconducting hardware, with all code and job identifiers retained. On the amplitudes side we verify, in exact rational/symbolic arithmetic: the canonical form of the ABHY associahedron against the bi-adjoint ϕ^3 amplitude; the six-term NMHV identity of the tree amplituhedron $\mathcal{A}(6,1,4)$ together with its facet structure (Gale evenness, 9 facets); the equality of the one-loop four-point box integrand with a pure dlog form; and the "maximally positive numerator" structure of the two-loop four-point integrand, whose double-box representation reproduces the positive part of the mutual condition $\langle ABCD\rangle$ exactly. On the emergence side we locate the transverse-field Ising transition variationally, extract the Ising central charge from entanglement ($c \approx 0.58$ vs. the exact 1/2 at N=12), and construct a six-tile holographic (HaPPY) network in which reconstruction depth, quantized information transfer, and a discrete Ryu–Takayanagi law are verified exactly. On hardware we measure: a CHSH violation $S = 2.7579$ with a no-signaling check; entanglement-wedge witnesses for a single holographic tile at 96% of ideal contrast after zero-noise extrapolation; and, for the 21-qubit six-tile network, bulk-operator localization—the same boundary avatar of a bulk operator reads +0.55 on the correct tile and +0.004 on the wrong one, a ∼130:1 contrast. We also document an instructive failure: zero-noise extrapolation diverges when circuit depth pushes raw signals to the noise floor, and the experiment's null witnesses correctly flagged the invalid values. This paper is a pedagogical verification study rather than a report of new theory; its contribution is a complete, reproducible chain from textbook nonlocality to research-frontier positive geometry, with each link executed rather than cited.

1 Introduction

Two research programs, developed largely independently, suggest that the basic furniture of physics—spacetime locality on one side, scattering processes on the other—may be derived

*Correspondence: epg@subnet345.com. Prepared by an AI agent swarm (a mix of OpenAI and Anthropic models) under the author's direction; see Acknowledgments.

rather than fundamental. In the holographic program, bulk spacetime emerges from patterns of entanglement among boundary degrees of freedom [1–4], with quantum error correction supplying the mechanism by which bulk information is redundantly and robustly encoded [5, 6]. In the amplitudes program, scattering amplitudes of specific quantum field theories are recovered as canonical differential forms of *positive geometries* [7]—the amplituhedron for planar $\mathcal{N} = 4$ super Yang–Mills [8, 9], the ABHY associahedron for bi-adjoint ϕ^3 theory [10], and cosmological polytopes for wavefunction coefficients [11]—so that locality and unitarity appear as consequences of positivity rather than axioms.

Both programs are unusually amenable to direct verification: their core claims are statements about finite-dimensional linear algebra, polytopes, rational functions, and few-qubit quantum states. This paper documents a complete execution of that observation. Rather than survey the literature, we *run* it: each foundational claim is reproduced as an exact computation (symbolic or rational arithmetic, deterministic statevector simulation) or as a measurement on IBM Quantum hardware, and the failures and corrections encountered along the way are reported alongside the successes.

We emphasize scope honestly at the outset. Nothing here is new theory. The hardware experiments are small-scale demonstrations, not competitive benchmarks. The value of the exercise, we believe, is threefold: it provides a reproducible pedagogical chain from elementary quantum information to research-frontier positive geometry; it demonstrates that several claims usually taken on authority (triangulation independence, the maximally positive two-loop numerator, quantized wedge information) can be checked end-to-end on commodity resources; and it supplies a case study in verification discipline, including one instructive mitigation failure caught by designed-in null tests.

2 Nonlocality without signaling

We prepare Bell pairs $|\Phi^+\rangle$ and evaluate the CHSH statistic $S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$ with measurement angles $(0, \pi/2)$ for one party and $(\pi/4, 3\pi/4)$ for the other. Local realism bounds $S \leq 2$; quantum mechanics attains Tsirelson's bound $2\sqrt{2} \approx 2.8284$.

On a noiseless simulator (2×10^4 shots per setting) we obtain $S = 2.8375 \pm 0.01$, consistent with Tsirelson's bound. On the 156-qubit IBM Heron processor `ibm_kingston` (job `d94sl25gc6cc73ffgo5g`), the four correlators measure $+0.6968$, -0.6681, $+0.6968$, $+0.6962$ against ideal ± 0.7071, giving

$$S_{\text{hw}} = 2.7579 \pm 0.010, \tag{1}$$

violating the classical bound by roughly 75 standard deviations. In the same runs, each party's marginal outcome distribution is independent of the remote setting to within 2×10^{-4}: the correlations are real and yet carry zero signaling capacity. This pair of facts—strong correlation, no channel—recurs as the organizing constraint of everything that follows.

3 Quantum error correction: from repetition to topology

Repetition code. The three-qubit bit-flip code encodes $\alpha|0\rangle + \beta|1\rangle \mapsto \alpha|000\rangle + \beta|111\rangle$ and measures the parities Z_0Z_1, Z_1Z_2 via ancillas. Because both codewords have identical parities, syndrome extraction acquires the *location* of a bit-flip while learning nothing about the encoded amplitudes; we verify that an encoded state with $P(1) = 0.750$ survives deterministic single-qubit attacks with corrected readout $P(1) = 0.746$–0.753 across all error positions. Under symmetric bit-flip noise of rate p per qubit, the measured logical error rate tracks the theoretical $3p^2 - 2p^3$ across $p \in [0.01, 0.55]$, crossing the break-even line at $p = 1/2$ exactly as predicted: redundancy helps only below threshold.

Surface code. We implement the distance-3 rotated surface code ("surface-17": 9 data, 8 ancilla qubits) with full two-round quantum syndrome extraction. Deterministic attack fingerprints are reproduced at 100% of shots: single X, Z, and Y errors light exactly their adjacent checks; the two-qubit string X_2X_5 lights only its *endpoints*; and the logical string $X_6X_7X_8$ produces a silent syndrome while flipping the logical qubit—the defining topological statement that the only invisible error spans the lattice. Under code-capacity bit-flip noise with a brute-force minimum-weight decoder, the logical error rate scales as $p_L \approx (13\text{–}15)\,p^2$ at small p, confirming that distance 3 removes all single faults.

The fair fight. Against symmetric X and Z noise (each at rate p, arbitrary stored state; 2×10^5 Monte Carlo trials per point, justified by Gottesman–Knill), the hierarchy inverts instructively: the repetition code is *worse than no encoding* (e.g. 0.0302 vs. 0.0200 at $p = 0.01$) because phase errors pass through it unseen, while surface-17 wins decisively below its $\sim$7% pseudo-threshold (0.0036 at $p = 0.01$) and loses above it (0.2256 vs. bare 0.1907 at $p = 0.10$). Full-spectrum protection at a constant-factor premium, plus a threshold, is the design pattern inherited by everything holographic below.

4 Emergent geometry from entanglement

We study the transverse-field Ising chain $H = -J\sum_i Z_iZ_{i+1} - h\sum_i X_i$.

Variational location of the transition. A 48-parameter hardware-efficient ansatz optimized by COBYLA reproduces exact ground-state energies for $N = 8$ across $h/J \in [0.2, 2.0]$, with the order parameter $\langle Z_iZ_{i+1}\rangle$ collapsing from 0.994 to 0.228 through the critical region. The variational error peaks at criticality (0.26 at $h = J$ vs. 0.03 deep in the ordered phase)—itself a diagnostic: critical states are the most entangled and hence hardest for shallow circuits.

Critical entanglement. For $N = 12$ (exact diagonalization), the half-chain entropy exhibits the well-known subtlety that the ordered phase contributes a constant $\ln 2$ of cat-state (GHZ) entropy. An infinitesimal symmetry-breaking field ($h_z = 10^{-3}$) collapses the cat and exposes the genuine critical peak ($S = 0.567$ near $h \approx 0.8$ at this size, vanishing in both phases). At $h = J$, fitting the Calabrese–Cardy law $S(\ell) = \frac{c}{6}\ln\big[\frac{2N}{\pi}\sin\frac{\pi\ell}{N}\big] + \text{const}$ yields $c = 0.58$, within finite-size expectations of the exact Ising value $c = 1/2$.

Entanglement as a ruler. Defining $d(i,j) = -\ln I(i,j)$ from two-site mutual information, the disordered phase ($h = 2J$) yields distances growing linearly with separation (1.6 per site: an ordinary ruler with a correlation length), while the critical chain yields logarithmic growth (scale-free connectivity). One coupling constant selects between two emergent geometries—the one-dimensional shadow of the tensor-network/holography correspondence [4, 12].

5 Positive geometry and scattering amplitudes

All results in this section are exact (symbolic or rational arithmetic).

The positive Grassmannian. For $\mathrm{Gr}(2,4)$ we verify the Plücker relation symbolically for a generic matrix, measure that only 2.14% of Gaussian-random 2-planes are totally positive, and confirm that the moment-curve parametrization renders every minor $t_j - t_i > 0$ automatically: positivity is equivalent to cyclic ordering. In a gauge fixing columns 1 and 3, total positivity reduces to a positive orthant with $p_{24} = ad + bc$ implied, and the cyclic product $p_{12}p_{23}p_{34}p_{14} = abcd$: the Parke–Taylor denominator *is* the canonical form of $\mathrm{Gr}_+(2,4)$.

Canonical forms and triangulation independence. We verify the recursive definition of canonical forms [7] bottom-up: interval residues ± 1; triangle edge-residues reproducing interval forms; and the unit square triangulated across both diagonals, with the spurious diagonal pole cancelling exactly in each sum and both triangulations agreeing.

The ABHY associahedron. Imposing only masslessness sum rules and the ABHY constants [10], a symbolic solve produces the $n = 5$ pentagon facets $\{x,\, y,\, c_{13} - x + y,\, c_{13} + c_{14} - x,\, c_{14} + c_{24} - y\}$. The five-term vertex sum (one planar Feynman diagram per vertex) combines over a common denominator equal exactly to the product of the five facets; residues on facets reproduce the boundary interval forms; vertex counts follow the Catalan numbers. Factorization—normally a theorem requiring locality and unitarity—is here the tautology that a boundary of a face is a face.

The tree amplituhedron $\mathcal{A}(6,1,4)$. With momentum twistors on a moment curve and Y interior, the two Britto–Cachazo–Feng–Witten (BCFW) triangulations of the NMHV hexagon polytope, $R_1 = [12345] + [12356] + [13456]$ and $R_2 = [23456] + [12456] + [12346]$, agree *exactly* in rational arithmetic ($R_1 - R_2 = 0$ at three random interior points)—the six-term NMHV identity as a statement about triangulations. Approaching the spurious hyperplane $\langle Y1235\rangle \to 0$, individual terms diverge as ± 86.8 while the sum holds at 1.5×10^{-4}. An exact same-side-of-hyperplane test classifies all $\binom{6}{4} = 15$ vertex 4-subsets into 9 facets— precisely the Gale-evenness set $\{i, i{+}1, j, j{+}1\}$, i.e. the physical propagator poles—and 6 interior seams.

One loop. In the single positive cell of the $n = 4$ MHV loop amplituhedron [9, 13], parametrized by $A = Z_1 {+} xZ_2 {+} wZ_3$, $B = -Z_1 {+} yZ_3 {+} zZ_4$, all physical brackets are monomials ($\langle AB12\rangle = wz$, $\langle AB23\rangle = z$, $\langle AB34\rangle = x$, $\langle AB14\rangle = xy$), the sign-flip sequence is $\{+, -, +\}$, and the box integrand divided by $d\log x \wedge d\log y \wedge d\log z \wedge d\log w$ equals 1 identically.

Two loops. For two loop lines the mutual condition evaluates to

$$\langle ABCD\rangle = (z_1 - z_2)(w_1x_2 - w_2x_1) + (x_1 - x_2)(y_1z_2 - y_2z_1), \tag{2}$$

and Monte Carlo sampling shows only 49.8% of independently valid loop pairs satisfy it: the geometry entangles the loops. Converting the physical double-box representation (s- and t-channel plus loop relabelings) to this chart and multiplying by $\langle ABCD\rangle$ yields the numerator

$$N = w_1x_2z_1 + w_2x_1z_2 + x_1y_1z_2 + x_2y_2z_1, \tag{3}$$

which is *exactly* the sum of the positive monomials of $\langle ABCD\rangle$—the "maximally positive numerator" phenomenon [13], here recovered from raw field theory. Two errors were caught by this cross-check during the work: an incorrect channel assignment in the double-box term list (diagnosed by a numerator with missing monomials) and the ordered-pair symmetrization convention (diagnosed by an overall factor of 2). We record these deliberately: the positivity structure functioned as a debugger for the physics input.

6 Holographic codes: exact results and hardware measurements

6.1 A single tile, exactly

The six-leg perfect tensor of the HaPPY code [6]—the encoding isometry of the [[5, 1, 3]] code, with the bulk leg entangled to a reference R—satisfies, in exact statevector arithmetic: all 20 three-leg bipartitions maximally entangled ($S = 3\ln 2$); mutual information $I(R{:}A) = 0$ for every boundary region with $|A| \leq 2$ and $I(R{:}A) = 2\ln 2$ for every $|A| \geq 3$ (the sharp entanglement-wedge transition); and weight-3 boundary representatives of the logical operators obtained by stabilizer multiplication, $\bar{X} = -(IYYIX)$ on qubits $\{2, 3, 5\}$ and $-(XIYYI)$ on $\{1, 3, 4\}$, verified to act identically on the bulk (operator pushing; cf. [5]).

6.2 A single tile on hardware

We prepared the tile state on `ibm_kingston` via a synthesized Clifford circuit (42 two-qubit gates after routing) and measured Pauli witnesses (Table 1): stabilizers, weight-6 logical correlators, three wedge avatars, and four null correlators between R and two-qubit regions. Raw (readout-mitigated) values average $+0.535$ for wedge witnesses. Adding zero-noise extrapolation (noise factors $1, 3, 5$; exponential fit), Pauli twirling, and dynamical decoupling raises the wedge mean to $+0.9558$ while the null mean *decreases* to 0.0046; two witnesses slightly exceed $+1$, the expected variance signature of an unbiased extrapolation. The asymmetry is the point: mitigation restores contrast that noise removed but cannot create correlations forbidden by the code—minority regions stay dark.

6.3 The six-tile network, exactly

Gluing five petal tiles to a central tile (6 bulk qubits, 5 bonds, 20 boundary qubits) yields, in exact contraction: reconstruction depth (a petal bulk qubit is recovered by a 3-qubit boundary arc; the central qubit requires 10 of 20); *quantized* information transfer, with $I(R{:}\mathrm{arc})$ taking only the values $\{0, 1, 2\} \times \ln 2$ (the intermediate shelf corresponding to a single logical Pauli inside the wedge); and a discrete Ryu–Takayanagi law, the arc entropies $S/\ln 2 = 1, 2, 2, 1, 2, 3, 3, 2, 3, 3$ for lengths 1–10 tracking the minimal bond cut, including the characteristic dips when the cut snaps from boundary legs to an internal bond.

6.4 The six-tile network on hardware

The 21-qubit network state is a stabilizer state; we derived its 21 generators numerically by pushing input Paulis through the petal isometries (choosing minimum-weight representatives from the Pauli decomposition of $W\sigma W^{\dagger}$) and synthesized a preparation circuit (148 two-qubit gates; 553 after routing) whose statevector matches the exact tensor-network state to overlap 1.0000000000.

Witness	Ideal	Raw	Mitigated
Stabilizer S_1	$+1$	$+0.9300$	$+0.9949$
Stabilizer S_2	$+1$	$+0.4235$	$+0.9996$
Stabilizer S_3	$+1$	$+0.4466$	$+0.9003$
Stabilizer S_4	$+1$	$+0.4221$	$+0.8897$
$X_R\bar{X}$ (weight 6)	$+1$	$+0.4450$	$+1.0504$
$Z_R\bar{Z}$ (weight 6)	$+1$	$+0.4036$	$+0.9668$
Bulk-X, wedge $\{2,3,5\}$	$+1$	$+0.4351$	$+1.0343$
Bulk-X, wedge $\{1,3,4\}$	$+1$	$+0.7707$	$+0.9411$
Bulk-Z, wedge $\{1,4,5\}$	$+1$	$+0.3993$	$+0.8921$
Null $X_RX_1X_2$	0	$+0.0015$	-0.0021
Null $Z_RZ_4Z_5$	0	-0.0146	-0.0146
Null $X_RZ_1Y_2$	0	$+0.0227$	$+0.0000$
Null $Z_RX_2X_3$	0	-0.0215	$+0.0017$
Mean wedge	$+1$	$+0.5350$	$+0.9558$
Mean |null|	0	0.0151	0.0046

Table 1: Single-tile witnesses on `ibm_kingston` (10^4 shots; raw job `d94tqhlgc6cc73ffi0eg`, mitigated job `d94ttgtgc6cc73ffi3p0`).

A first hardware attempt with zero-noise extrapolation (job `d94u25nu62ks7396fhlg`) *diverged*: null witnesses returned values of order $\pm 4\times 10^9$, impossible for Pauli expectations. The cause is generic and worth recording: at $\sim$550 two-qubit gates, several raw signals sit at the noise floor already at unit noise, and exponential extrapolation through noise-floor points amplifies fit noise without bound. The null tests functioned exactly as designed, flagging the entire dataset—including superficially plausible entries—as artifacts.

The clean rerun (job `d94u424ql68s73c9vtig`; twirling, readout mitigation, dynamical decoupling, no extrapolation) gives physical values throughout (Table 2). The headline is the localization contrast:

$$\underbrace{+0.5519}_{\text{bulk-}X\text{ avatar on petal 1}} \qquad \text{vs.} \qquad \underbrace{+0.0042}_{\text{identical operator on petal 2}} \tag{4}$$

a $\sim$130:1 ratio at $\sim 55\sigma$: the same boundary operator reads the bulk qubit through the correct tile and reads exactly nothing through the wrong one. The remaining wedge witnesses survive at reduced contrast (+0.099, +0.065; both above the null floor), and the per-petal encoding checks (0.07–0.24) quantify how badly 553 noisy gates bruise the state. A bulk point of a six-tile emergent geometry was located—and localized—on physical hardware, at the edge of what unmitigated depth allows.

7 Discussion

Three observations organize what the exercise taught us.

One trick, three scales. The repetition code protects a bit by majority; the surface code protects a qubit by homology; the holographic code protects a bulk point by wedge structure. In each case information is stored in a global property that no sub-threshold local process

Witness	Ideal	Measured
Image check, petal 1	+1	+0.0884
Image check, petal 2	+1	+0.0692
Image check, petal 3	+1	+0.1218
Image check, petal 4	+1	+0.1979
Image check, petal 5	+1	+0.2424
Wedge: bulk-X avatar A, petal 1	+1	+0.5519
Wedge: bulk-X avatar B, petal 1	+1	+0.0993
Wedge: bulk-Z avatar, petal 1	+1	+0.0648
Null: same bulk-X avatar, petal 2	0	+0.0042
Null: X_RXX, petal-1 pair	0	+0.0088
Null: Z_RZZ, petal-3 pair	0	−0.0109

Table 2: Six-tile flower network on `ibm_kingston` (10^4 shots, job `d94u424ql68s73c9vtig`; twirling + readout mitigation + dynamical decoupling, no extrapolation).

reaches, and in each case the protection pattern *is* a geometry (a line, a torus, a hyperbolic disk). Our hardware results trace the same arc operationally: near-perfect contrast at 2 qubits, 96% mitigated contrast at 6, a surviving-but-bruised 55σ signal at 21. The scaling wall is exponential in circuit depth, and the known way through it is the error correction of Section 3—which is also the mechanism under study. The experiments are, in a precise sense, error-correction-limited studies of error correction.

Positivity as output specification. At tree level, positivity fixed pole locations; at one loop, it reduced the box to a pure dlog form; at two loops it fixed the *numerator* exactly. Twice during this work, discrepancies between a hand-supplied physics formula and the positive-geometry prediction turned out to be errors in the former. Whatever positive geometry ultimately is, it is at minimum an unusually effective consistency engine for perturbative field theory.

No levers. Every prospective "exploit" the underlying emergent-spacetime narrative might suggest was tested and closed by the physics itself: entanglement carries no signal (Section 2); minority wedges carry no information (Sections 6); adjacency in kinematic space is descriptive, with positivity *enforcing* rather than circumventing locality and causality (Section 5). The constraints are not guardrails around the structure; they are the structure.

Limitations. All models studied are toys: the amplituhedron addresses planar $\mathcal{N} = 4$ super Yang–Mills; the associahedron, bi-adjoint ϕ^3; the HaPPY network, a static AdS-like geometry. Whether our universe employs any of these mechanisms is open; de Sitter holography and the emergence of *time* have no comparable toy models at all. The hardware demonstrations use few qubits, favorable witnesses, and a free-tier device; they demonstrate structure, not capability. The recent two-loop geometry literature we verified against [14] is young and single-group.

8 Methods

All computations used Python 3.13 (NumPy/SciPy/SymPy) and Qiskit 2.5 with Aer for exact simulation; hardware runs used IBM Quantum (`ibm_kingston`, Heron, 156 qubits) via the EstimatorV2/SamplerV2 primitives with 10^4 shots per observable, and (where stated) resilience options: twirled readout mitigation, Pauli gate twirling, XY4 dynamical decoupling, and zero-noise extrapolation with gate folding. Exact claims use symbolic or rational arithmetic throughout; no floating-point equality is asserted anywhere in Section 5. Stabilizer-state preparation circuits were synthesized from explicit generator lists and verified against exact statevectors before submission. All literature-derived formulas (one-loop cell parametrization, two-loop mutual condition and numerator structure, amplituhedron region definitions) were checked against their sources [9, 13, 14] rather than reproduced from memory. Complete source code (thirteen laboratory scripts), an annotated 18-paper reference corpus with evidence tiers, and all IBM job identifiers are retained by the author.

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